

Team Pairadox

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Explainable BGP Origin and Subprefix Anomaly Monitor

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Problem Description

Detect and explain suspicious BGP route announcements.

- Unexpected origin AS changes may indicate routing hijacks or configuration errors. - New subprefix announcements may redirect traffic to a different network.

- Our monitor will combine routing history and RPKI validation to flag anomalies.

- Each alert will show the affected prefix, observed origin and supporting evidence.

Background

BGP routing, origin changes and subprefix announcements.

- BGP exchanges reachability information between autonomous systems (ASes). [1]

- An origin ASN identifies the network originating a route. New origins require context. - A subprefix is a more-specific address range within an existing prefix.

- RPKI validation compares prefix, ASN and length with validated authorizations. [2-4] - States are Valid, Invalid and NotFound. An anomaly alone does not prove a hijack.

Literature & RFC Review

Relevant standards and data sources for the proposed monitor.

- RFC 4271 defines BGP and guides route-update parsing. [1]

- RFC 6811 and RFC 8481 define and clarify origin validation. [2,3]

- RFC 9582 specifies the Route Origin Authorization (ROA) profile. [4]

- CAIDA BGPStream supports historical and live routing-data access. [5]

- We propose explanations that combine validation results with route-history context.

Proposed Solution

A passive monitor with clear, evidence-based alerts.

- Ingest recorded BGP announcements and withdrawals, then explore a live feed.

- Track active routes per collector and peer, historical origins and covering prefixes. - Flag new origins, multiple-origin events and unexpected subprefix announcements. - Enrich candidates with RPKI state and record the freshness of supporting data.

- Explain triggered rules in a dashboard. Proposed stack: Python, SQLite, Streamlit.

Optimization Proposed

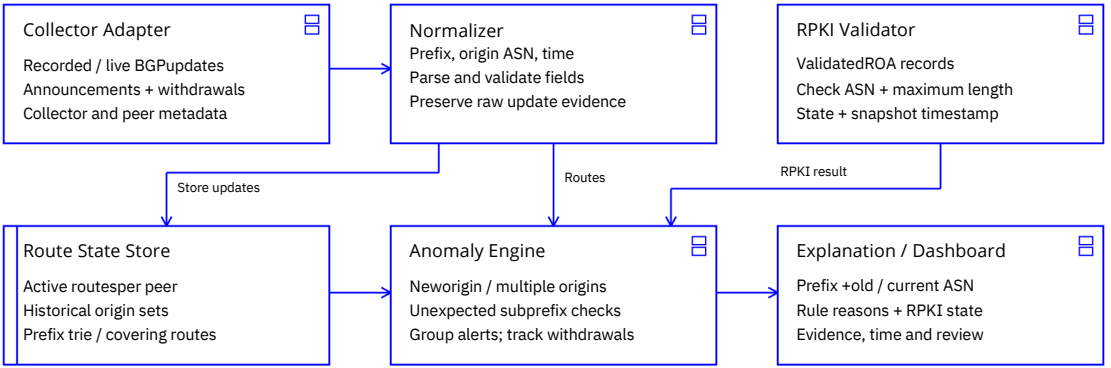
Enhancements over a simple origin-change detector.

Baseline	Proposed enhancement
Exact-prefix matching	Prefix trie for covering-prefix checks
Repeated update alerts	Group incidents and track withdrawals
Origin change alone	Add RPKI and historical context
Generic warnings	Show rule reasons and raw evidence

Solution Architecture & Design

Proposed components and their responsibilities.

UML component view: collection, validation, detection and review.

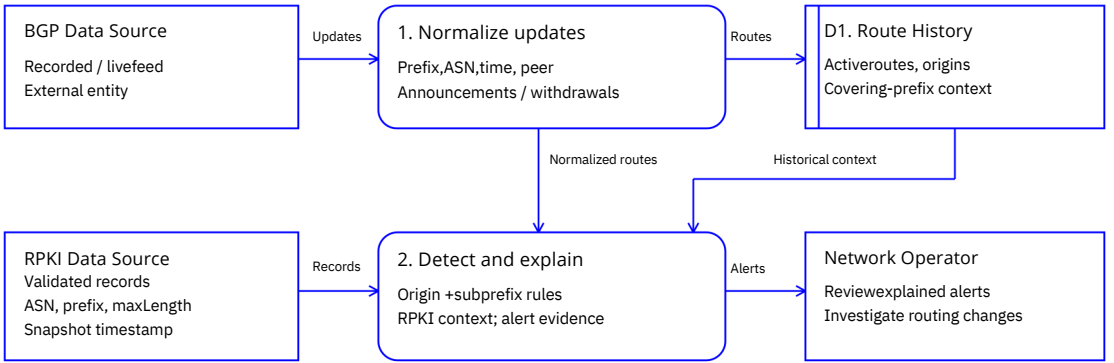


Proposed stack: Python + SQLite + Streamlit. Alerts support investigation.

Data Flow Diagram

Data flow from routing observations to operator review.

DFD: labeled data flows, processes, stores and external entities.



Alert payload: prefix, previous/current ASN, triggered rules, RPKI state and time.

Deployment Timeline

Proposed development milestones

- Phase 1: Review standards, select replay data and define labeled test scenarios.
- Phase 2: Snehali Dey develops detection and backend; Aratreeka iPahari ntegrates data and RPKI.
- Phase 3: Integrate explanations and dashboard, with technical review by Dr. Singh.
- Phase 4: Test benign changes and anomalies; measure precision, recall and delay.
- Deliver a reproducible local demo. Roles and phase durations remain proposed.

Bibliography I

[1] Rekhter, Li and Hares. RFC 4271: A Border Gateway Protocol 4. 2006.
https://www.rfc-editor.org/info/rfc4271/

[2] Mohapatra et al. RFC 6811: BGP Prefix Origin Validation. 2013.
https://www.rfc-editor.org/info/rfc6811/

[3] Bush. RFC 8481: Clarifications to BGP Origin Validation. 2018.
https://www.rfc-editor.org/info/rfc8481/

[4] RFC 9582: A Profile for Route Origin Authorizations (ROAs). 2024.
https://www.rfc-editor.org/info/rfc9582/

[5] CAIDA. BGPStream: Data & Meta-data Access.
https://bgpstream.caida.org/docs/overview/data-access

Thank You for Your Attention!